

B1 Fluids, 2021 — Model Solutions

Question 1 — Stokes flow between hinged plates

Problem. High vs low Re; biharmonic streamfunction; viscous flow between two long, wide plates hinged at $r = 0$, opening at constant $\dot{\alpha}$; verify the quoted ψ , find breakdown scales, flux, force scaling, and discuss the scallop theorem.

(a) High vs low Reynolds number

$\text{Re} = UL/\nu$ measures inertia/viscous force ratio. High Re ($\gg 1$): inertia dominates; viscosity confined to thin boundary layers $\delta \sim L/\sqrt{\text{Re}}$; turbulence; form drag. Low Re ($\ll 1$): linear Stokes equation $\nabla p = \mu \nabla^2 \mathbf{v}$; time-reversible, instantaneous, drag $\propto U$.

Swimming human: $L \sim 1 \text{ m}$, $U \sim 1 \text{ m s}^{-1}$, $\nu = 10^{-6} \text{ m}^2 \text{ s}^{-1}$:

$$\text{Re} \sim \frac{1 \cdot 1}{10^{-6}} = 10^6.$$

$\text{Re} \sim 10^6 \gg 1$: high-Re regime.

(b) Streamfunction and biharmonic equation

Polar incompressibility:

$$\nabla \cdot \mathbf{v} = \frac{1}{r} \frac{\partial(rv_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta}.$$

Substitute $v_r = (1/r)\partial_\theta \psi$, $v_\theta = -\partial_r \psi$:

$$\nabla \cdot \mathbf{v} = \frac{1}{r} \frac{\partial}{\partial r} \left(\frac{\partial \psi}{\partial \theta} \right) - \frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{\partial \psi}{\partial r} \right).$$

Equality of mixed partials:

$$\nabla \cdot \mathbf{v} = 0.$$

Vorticity in 2D: $\boldsymbol{\omega} = \omega \hat{\mathbf{z}}$, $\omega = -\nabla^2 \psi$. At low Re, drop inertia in NS:

$$\nabla p = \mu \nabla^2 \mathbf{v}. \tag{1}$$

Take curl of (1):

$$0 = \mu \nabla^2 \boldsymbol{\omega}.$$

Substitute $\omega = -\nabla^2 \psi$:

$\nabla^{22} \psi = \nabla^4 \psi = 0.$

(2)

(c) Boundary conditions and verification

BCs. No-slip on the fixed plate ($\theta = 0$):

$$v_r(r, 0) = 0, \quad v_\theta(r, 0) = 0 \implies \left. \frac{\partial \psi}{\partial \theta} \right|_0 = 0, \quad \left. \frac{\partial \psi}{\partial r} \right|_0 = 0.$$

Rigid rotation of upper plate ($\theta = \alpha$): $\mathbf{v}_{\text{plate}} = r\dot{\alpha}\hat{\boldsymbol{\theta}}$:

$$v_r(r, \alpha) = 0, \quad v_\theta(r, \alpha) = r\dot{\alpha} \implies \left. \frac{\partial \psi}{\partial \theta} \right|_\alpha = 0, \quad \left. \frac{\partial \psi}{\partial r} \right|_\alpha = -r\dot{\alpha}.$$

Ansatz. BC linear in r forces $n = 2$:

$$\psi(r, \theta) = r^2 f(\theta), \quad f(\theta) = A \cos 2\theta + B \sin 2\theta + C\theta \cos 2\theta + D\theta \sin 2\theta.$$

Reduce BCs by dividing r^2 and $2r$:

$$f(0) = 0, \quad f'(0) = 0, \quad f(\alpha) = -\frac{1}{2}\dot{\alpha}, \quad f'(\alpha) = 0.$$

Apply $f(0) = 0$:

$$A = 0.$$

Differentiate f :

$$f'(\theta) = -2A \sin 2\theta + 2B \cos 2\theta + C(\cos 2\theta - 2\theta \sin 2\theta) + D(\sin 2\theta + 2\theta \cos 2\theta).$$

Apply $f'(0) = 0$:

$$2B + C = 0 \implies C = -2B.$$

Reduced f :

$$f(\theta) = B \sin 2\theta - 2B\theta \cos 2\theta + D\theta \sin 2\theta.$$

Reduced f' :

$$f'(\theta) = 4B\theta \sin 2\theta + D(\sin 2\theta + 2\theta \cos 2\theta).$$

Write $s = \sin 2\alpha$, $c = \cos 2\alpha$. The two upper-plate conditions become

$$4B\alpha s + D(s + 2\alpha c) = 0,$$

$$B(s - 2\alpha c) + D\alpha s = -\frac{1}{2}\dot{\alpha}.$$

Determinant:

$$\Delta = 4\alpha^2 s^2 - (s + 2\alpha c)(s - 2\alpha c).$$

Cramer for B :

$$B = \frac{\dot{\alpha}(s + 2\alpha c)}{2\Delta}.$$

Cramer for D :

$$D = -\frac{2\alpha s \dot{\alpha}}{\Delta}.$$

Substitute back into ψ :

$$\boxed{\psi = -\frac{\dot{\alpha} r^2}{2} \frac{(4\alpha \sin 2\alpha)(\theta \sin 2\theta) - (\sin 2\alpha + 2\alpha \cos 2\alpha)(\sin 2\theta - 2\theta \cos 2\theta)}{4\alpha^2 \sin^2 2\alpha - (\sin 2\alpha + 2\alpha \cos 2\alpha)(\sin 2\alpha - 2\alpha \cos 2\alpha)}} \quad (3)$$

which matches the quoted formula with $\Omega = \dot{\alpha}$.

Breakdown scales. Local Reynolds number $\text{Re}_{\text{loc}} = Ur/\nu$, $U \sim r\dot{\alpha}$:

$$\text{Re}_{\text{loc}} \sim \frac{r^2 \dot{\alpha}}{\nu} \lesssim 1.$$

Solve for r :

$$l_1 = \sqrt{\nu/\dot{\alpha}}.$$

End-effects of finite plate length:

$$l_2 = L.$$

$$\boxed{l_1 = \sqrt{\nu/\dot{\alpha}}, \quad l_2 = L.}$$

(d) Volume flux through cylinder of radius r

Flux per unit length:

$$Q(r) = \int_0^\alpha v_r r \, d\theta = \int_0^\alpha \frac{\partial \psi}{\partial \theta} d\theta = \psi(r, \alpha) - \psi(r, 0).$$

Use $f(0) = 0$, $f(\alpha) = -\frac{1}{2}\dot{\alpha}$:

$$\boxed{Q(r) = -\frac{1}{2}\dot{\alpha} r^2.} \tag{4}$$

Geometric meaning: rate of area swept inside radius r as α grows is $\frac{1}{2}r^2\dot{\alpha}$, so fluid is drawn in across the cylinder at exactly that rate (mass conservation).

(e) Free hinge: pressure, force, scallop theorem

Pressure. With ψ known, $\mathbf{v}$ is known; integrate Stokes (1):

$$\nabla p = \mu \nabla^2 \mathbf{v} = -\mu \nabla \times (\omega \hat{\mathbf{z}}).$$

Line-integrate from a reference point; fix the constant by matching ambient pressure at $r = L$.

Force scaling. Stokes flow is linear in boundary velocity, so $p \sim \mu\dot{\alpha}$ (independent of r to leading order); plate length L :

$$F \sim p \cdot L.$$

$$\boxed{F \propto \mu L \dot{\alpha}.}$$

Scallop theorem. Periodic $\alpha(t)$, period T . Net impulse:

$$\mathcal{I} = \oint F \, dt \propto \mu L \oint \dot{\alpha} \, dt = \mu L [\alpha(T) - \alpha(0)].$$

Periodicity:

$$\mathcal{I} = 0.$$

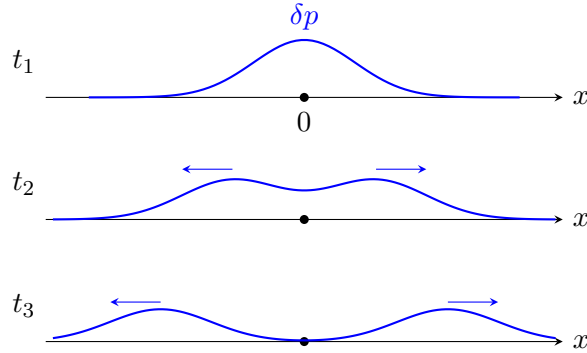
A reciprocal one-DOF gait produces no net displacement (Purcell 1977). Cyclic opening and closing of a hinged scallop cannot generate propulsion at low Re ; non-reciprocal gaits (e.g. rotating helical flagella) are required.

Question 2 — Sound waves in a moving medium

Problem. Mechanism of sound; convected wave equation in a uniform stream $U_0\hat{\mathbf{x}}$; dispersion; observers up- and downstream; supersonic and transonic cases.

(a) Mechanism of sound propagation

A local compression raises p above ambient. Pressure gradient $-\partial_x p$ accelerates neighbouring fluid outward (Newton: $\rho_0 \partial_t v = -\partial_x p$). Mass piles into next layer; that layer compresses; the bump translates. Speed: inertia/compressibility balance $c_0 = \sqrt{\partial p / \partial \rho|_{\rho_0}}$. Pulse splits into left- and right-going branches (d'Alembert).



(b) Linearisation: convected wave equation

Write $\rho = \rho_0 + \rho_1$, $p = p_0 + p_1$, $v = U_0 + v_1$.

Continuity:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0.$$

Linearise:

$$D\rho_1 + \rho_0 \frac{\partial v_1}{\partial x} = 0, \quad D \equiv \frac{\partial}{\partial t} + U_0 \frac{\partial}{\partial x}. \quad (5)$$

1-D inviscid Euler:

$$\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = 0.$$

Linearise:

$$Dv_1 = -\frac{1}{\rho_0} \frac{\partial p_1}{\partial x}. \quad (6)$$

Polytropic state: linearise $p = p_0(\rho/\rho_0)^\gamma$:

$$p_1 = c_0^2 \rho_1, \quad c_0^2 = \left. \frac{dp}{d\rho} \right|_{\rho_0} = \frac{\gamma p_0}{\rho_0}. \quad (7)$$

Apply D to (5):

$$D^2 \rho_1 + \rho_0 D \left(\frac{\partial v_1}{\partial x} \right) = 0.$$

Commute D and ∂_x ; substitute (6):

$$\rho_0 D \left(\frac{\partial v_1}{\partial x} \right) = \rho_0 \frac{\partial (Dv_1)}{\partial x} = -\frac{\partial^2 p_1}{\partial x^2}.$$

Substitute (7):

$$\rho_0 D\left(\frac{\partial v_1}{\partial x}\right) = -c_0^2 \frac{\partial^2 \rho_1}{\partial x^2}.$$

Hence

$$\boxed{\left(\frac{\partial}{\partial t} + U_0 \frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + U_0 \frac{\partial}{\partial x}\right) \rho_1 = c_0^2 \frac{\partial^2 \rho_1}{\partial x^2}.} \quad (8)$$

(c) Dispersion and snapshot at $U_0 = 0$

Plane wave $\rho_1 = \rho_0 A e^{i(kx - \omega t)}$ in (8):

$$(-i\omega + iU_0 k)^2 = -c_0^2 k^2.$$

Square root:

$$\omega - U_0 k = \pm c_0 k.$$

$$\boxed{\omega = (U_0 \pm c_0)k.} \quad (9)$$

Wavenumbers at fixed ω :

$$k_{\pm} = \frac{\omega}{U_0 \pm c_0}.$$

Snapshot $U_0 = 0$, $\omega t = 2n\pi$. Outgoing solution:

$$\rho_1 = \rho_0 A \cos[\omega(|x|/c_0 - t)].$$

At $\omega t = 2n\pi$:

$$\rho_1 = \rho_0 A \cos(k_0 |x|), \quad k_0 = \omega/c_0.$$

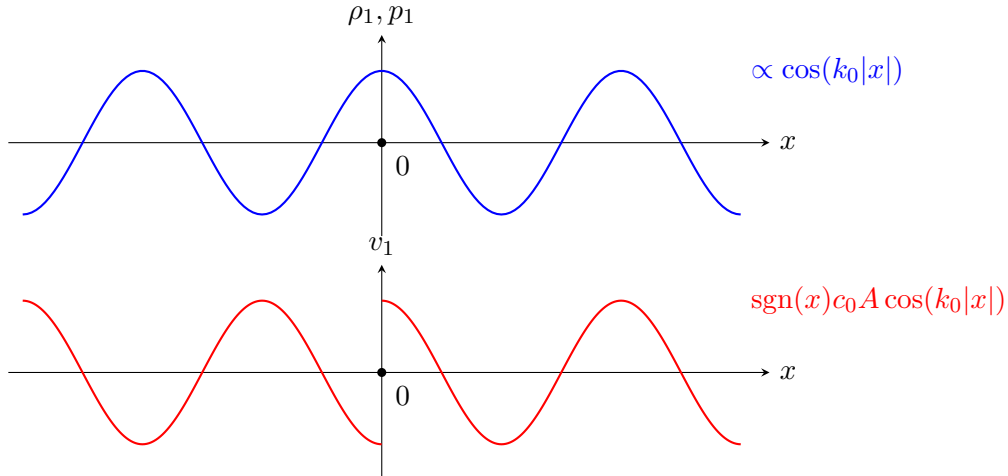
Use (7):

$$p_1 = c_0^2 \rho_1.$$

Use (6) with $U_0 = 0$ and outgoing waves $\propto e^{i(k|x| - \omega t)}$:

$$v_1 = \text{sgn}(x) c_0 A \cos(k_0 |x|).$$

ρ_1, p_1 even in x ; v_1 odd (radiation away from source).



(d) Two observers; supersonic; $U_0 \rightarrow c_0$

Wavelengths:

$$\lambda_{\pm} = \frac{2\pi(U_0 \pm c_0)}{\omega} \quad (\text{left-going uses } |U_0 - c_0|).$$

Subsonic $U_0 < c_0$. Both observers see frequency ω (steady forcing in lab frame). Downstream ($x = +L$): wavelength $\lambda_+ = 2\pi(U_0 + c_0)/\omega$ (stretched). Upstream ($x = -L$): wavelength $|\lambda_-| = 2\pi(c_0 - U_0)/\omega$ (compressed). Ratio of wavecount in window l :

$$\frac{N_-}{N_+} = \frac{U_0 + c_0}{c_0 - U_0}.$$

Supersonic $U_0 > c_0$. Both $k_{\pm} > 0$; both branches advected downstream. At $x = +L$ a beat pattern of two waves with $k_{\pm} = \omega/(U_0 \pm c_0)$. At $x = -L$: nothing — no characteristic reaches $x = -L$ from $x = 0$.

Limit $U_0 \rightarrow c_0^-$. $\lambda_- \rightarrow 0$, lab phase speed $c_0 - U_0 \rightarrow 0$; upstream wave is frozen and energy density diverges under finite forcing. Linear theory breaks down: nonlinear steepening (shock) and dissipation must be reintroduced — the transonic singularity.

Question 3 — Lubrication theory for a sessile drop

Problem. Thin axisymmetric drop, density ρ , viscosity μ , surface tension γ , gravity g . Justify lubrication NS, derive $p(r, z, t)$ and the parabolic $\mathbf{v}$. Volume flux gives the thin-film PDE. Steady state with $\kappa \approx -\partial_r^2 h$, contact angle θ_c , regularity at $r = 0$, volume V .

(a) Lubrication, pressure, horizontal velocity

Let H, L be vertical/horizontal scales, $\epsilon \equiv H/L \ll 1$. Continuity:

$$\frac{U}{L} \sim \frac{W}{H} \implies W \sim \epsilon U \ll U.$$

Reduced Reynolds (inertia/viscous):

$$\frac{\rho U^2/L}{\mu U/H^2} = \text{Re } \epsilon^2 \ll 1.$$

Streamwise viscous term suppressed by ϵ^2 . Hence

$$\boxed{0 \approx \mu \frac{\partial^2 \mathbf{v}}{\partial z^2} - \nabla p - \rho g \hat{\mathbf{z}}.} \quad (10)$$

Pressure. z -component: $\partial_z p = -\rho g$. Integrate:

$$p(r, z, t) = -\rho g z + f(r, t).$$

Apply Young–Laplace at $z = h$: $p(r, h) = p_0 - \gamma \kappa$:

$$f = p_0 + \rho g h - \gamma \kappa.$$

Hence

$$\boxed{p(r, z, t) = p_0 + \rho g(h - z) - \gamma \kappa(r, t).} \quad (11)$$

Horizontal velocity. Radial component of (10):

$$\mu \frac{\partial^2 v_r}{\partial z^2} = \frac{\partial p}{\partial r}.$$

Differentiate (11):

$$\frac{\partial p}{\partial r} = \rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial(-\kappa)}{\partial r}.$$

Write driving term:

$$G(r, t) \equiv \rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \quad (\text{sign convention of question}).$$

Integrate once with $\partial_z v_r|_{z=h} = 0$ (no surface stress):

$$\mu \frac{\partial v_r}{\partial z} = G(z - h).$$

Integrate again with $v_r(0) = 0$ (no-slip):

$$\mu v_r = G\left(\frac{1}{2}z^2 - hz\right) = \frac{1}{2}G z(z - 2h).$$

Vector form:

$$\boxed{\mathbf{v} \approx \left[\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right] \frac{z(z - 2h)}{2\mu} \hat{\mathbf{r}}.} \quad (12)$$

(b) Volume flux and the thin-film equation

Flux through cylinder of radius r :

$$Q(r, t) = 2\pi r \int_0^h v_r \, dz.$$

Substitute (12):

$$Q = 2\pi r \frac{1}{2\mu} \left(\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right) \int_0^h z(z - 2h) \, dz.$$

Evaluate inner integral:

$$\int_0^h z(z - 2h) \, dz = \frac{1}{3}h^3 - h^3 = -\frac{2}{3}h^3.$$

Hence

$$Q(r, t) = -\frac{2\pi r h^3}{3\mu} \left(\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right).$$

Annular mass balance, $r \leq r' \leq r + dr$:

$$\frac{\partial}{\partial t} (2\pi r h \, dr) = -\frac{\partial Q}{\partial r} dr.$$

Cancel $2\pi dr$ and substitute Q :

$$\boxed{\frac{\partial h}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left[r \frac{h^3}{3\mu} \left(\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right) \right].} \quad (13)$$

(c) Steady drop shape

Steady $\partial_t h = 0$: Q independent of r . Boundary $h(R) = 0$ forces $Q \equiv 0$. With $h > 0$:

$$\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} = 0.$$

Integrate in r :

$$\rho g h + \gamma \kappa = K \quad (\text{const}).$$

Use $\kappa \approx -\partial_r^2 h$:

$$\gamma \frac{\partial^2 h}{\partial r^2} = \rho g h - K.$$

Define capillary length:

$$l \equiv \sqrt{\gamma/(\rho g)}.$$

ODE:

$$\frac{\partial^2 h}{\partial r^2} - \frac{1}{l^2} h = -\frac{K}{\gamma}.$$

General bounded solution (regular at $r = 0$ requires sinh coefficient zero):

$$h(r) = A_1 \cosh(r/l) + h_0, \quad h_0 \equiv Kl^2/\gamma.$$

Apply $h(R) = 0$:

$$h_0 = -A_1 \cosh(R/l).$$

Substitute:

$$h(r) = -A_1 [\cosh(R/l) - \cosh(r/l)].$$

Set $\alpha \equiv -A_1/l > 0$:

$$h(r) = \alpha l [\cosh(R/l) - \cosh(r/l)]. \quad (14)$$

Apply $\partial_r h(R) = -\tan \theta_c$:

$$-\alpha \sinh(R/l) = -\tan \theta_c.$$

Solve:

$$\alpha = \frac{\tan \theta_c}{\sinh(R/l)}.$$

Volume:

$$V = \int_0^R 2\pi r h(r) dr.$$

Substitute (14) and let $\xi = r/l$:

$$V = 2\pi \alpha l^3 \int_0^{R/l} [\cosh(R/l) - \cosh \xi] \xi d\xi.$$

Divide by V :

$$1 = \frac{2\pi l^3 \alpha}{V} \int_0^{R/l} [\cosh(R/l) - \cosh \xi] \xi d\xi. \quad (15)$$

Force balances. Bulk: $\rho g h$ (gravity hydrostatic) vs $-\gamma \partial_r^2 h$ (capillary Laplace pressure); crossover at the capillary length l . $R \ll l$: surface tension dominates, drop is a spherical cap. $R \gg l$: gravity dominates, drop is a flat puddle of thickness $\sim 2l \sin(\theta_c/2)$. Edge: Young's equation $\gamma_{SV} = \gamma_{SL} + \gamma \cos \theta_c$ fixes θ_c , hence α . Global: (15) ties R to V .

Question 4 — Complex potential and Joukowski mapping

Problem. Why simple shear has no $w(z)$. Streamfunctions of $w = \beta z^2$ and $w = \beta z^2 e^{i\alpha}$; dyed disc area. Cylinder of radius a in the second flow: find $w = Az^2 + B/z^2$, surface speed, net force. Joukowski map to ellipse: $W(Z)$ and stagnation points.

(a) Why simple shear has no $w(z)$

$\mathbf{v} = (\beta y, 0)$. Divergence:

$$\nabla \cdot \mathbf{v} = 0.$$

Curl:

$$(\nabla \times \mathbf{v})_z = -\beta \neq 0.$$

$w(z) = \phi + i\psi$ requires $\mathbf{v} = \nabla\phi$ (irrotational) and $\nabla \cdot \mathbf{v} = 0$. Simple shear has uniform vorticity $-\beta$, so no ϕ exists, hence no $w(z)$.

(b) Streamfunctions and the dyed disc

(i) $w = \beta z^2$. Expand:

$$w = \beta(x^2 - y^2) + 2i\beta xy.$$

Hence

$$\boxed{\phi = \beta(x^2 - y^2), \quad \psi = 2\beta xy.}$$

Streamlines: $xy = \text{const}$ (rectangular hyperbolae); axes are $\psi = 0$. Pure straining flow with stagnation at origin, principal axes along x and y .

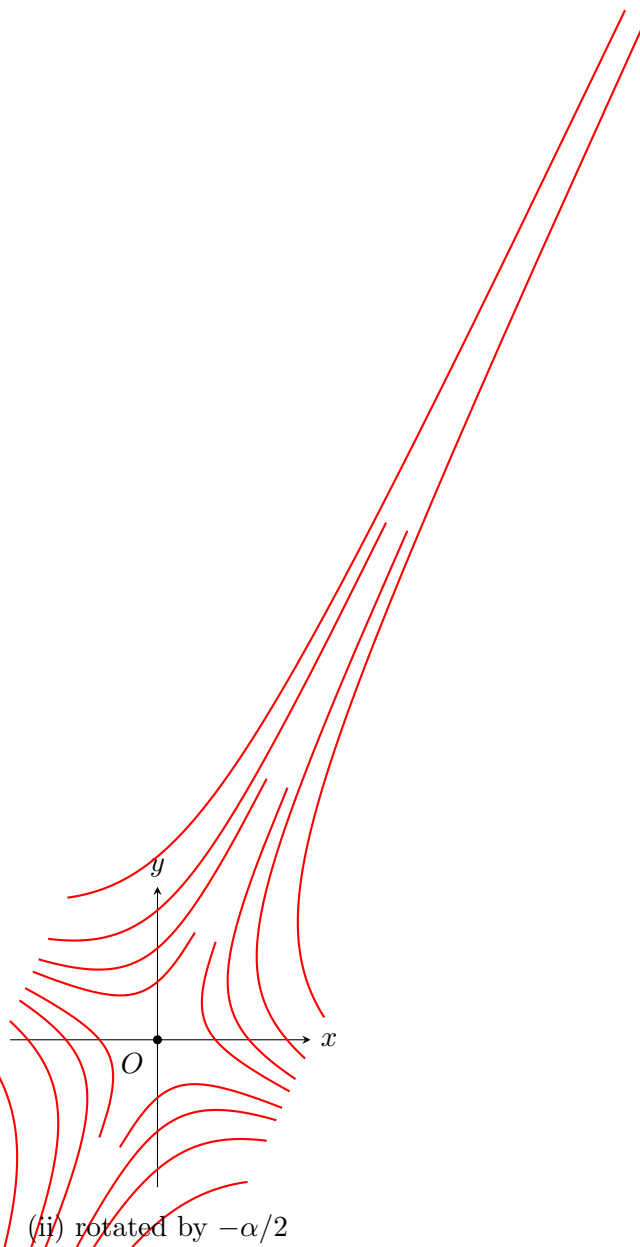
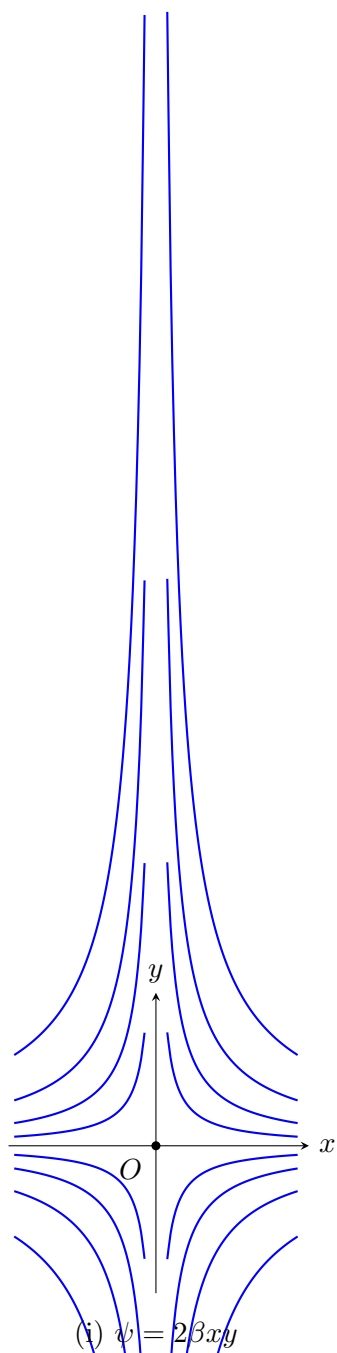
(ii) $w = \beta z^2 e^{i\alpha}$. Polar $z = re^{i\theta}$:

$$w = \beta r^2 e^{i(2\theta + \alpha)}.$$

Imaginary part:

$$\boxed{\psi = \beta r^2 \sin(2\theta + \alpha).}$$

Streamlines are the same hyperbolae rotated by $-\alpha/2$.



Dyed disc area. 2D incompressible flow conserves material area:

$$\boxed{\text{Area}(t) = \pi R^2 \text{ for all } t.}$$

The disc deforms into an ellipse: stretches by $e^{2\beta t}$ along the extensional axis, compresses by $e^{-2\beta t}$ along the orthogonal axis. Product unity.

(c) Cylinder in the strain field

Find A, B . Far field $w \sim \beta z^2 e^{i\alpha}$:

$$A = \beta e^{i\alpha}.$$

On $z = ae^{i\theta}$:

$$w = Aa^2 e^{2i\theta} + \frac{B}{a^2} e^{-2i\theta}.$$

Impermeability requires $\text{Im } w = \text{const}$ on $|z| = a$ for all θ . Write $A = \beta(\cos \alpha + i \sin \alpha)$, $B = B_R + iB_I$:

$$\text{Im } w = \beta a^2 \sin(2\theta + \alpha) + \frac{1}{a^2} [B_I \cos 2\theta - B_R \sin 2\theta].$$

Match $\cos 2\theta$ coefficient:

$$\beta a^2 \sin \alpha + B_I/a^2 = 0 \implies B_I = -\beta a^4 \sin \alpha.$$

Match $\sin 2\theta$ coefficient:

$$\beta a^2 \cos \alpha - B_R/a^2 = 0 \implies B_R = \beta a^4 \cos \alpha.$$

Combine:

$$B = \beta a^4 e^{-i\alpha}.$$

Hence

$$\boxed{w(z) = \beta e^{i\alpha} z^2 + \frac{\beta a^4 e^{-i\alpha}}{z^2}.} \quad (16)$$

Surface speed. Differentiate:

$$\frac{dw}{dz} = 2\beta e^{i\alpha} z - \frac{2\beta a^4 e^{-i\alpha}}{z^3}.$$

Set $z = ae^{i\theta}$:

$$\frac{dw}{dz} = 2\beta a e^{i(\alpha+\theta)} - 2\beta a e^{-i(\alpha+3\theta)}.$$

Factor $e^{-i\theta}$:

$$\frac{dw}{dz} = 2\beta a e^{-i\theta} \left[e^{i(\alpha+2\theta)} - e^{-i(\alpha+2\theta)} \right].$$

Use $e^{i\phi} - e^{-i\phi} = 2i \sin \phi$:

$$\frac{dw}{dz} = 4i\beta a e^{-i\theta} \sin(2\theta + \alpha).$$

Modulus:

$$\boxed{v = \left| \frac{dw}{dz} \right| = 4a\beta |\sin(2\theta + \alpha)|.} \quad (17)$$

Force on a surface element; net force. Inviscid: only pressure stress. Bernoulli (steady, irrotational):

$$p(\theta) = p_\infty - \frac{1}{2}\rho v^2(\theta).$$

Element force per unit length: $d\mathbf{F} = -p(\theta) \hat{\mathbf{r}} a d\theta$ (pressure pushes inward along $-\hat{\mathbf{r}}$). Net force per unit length:

$$\mathbf{F} = - \int_0^{2\pi} [p_\infty - 8\rho a^2 \beta^2 \sin^2(2\theta + \alpha)] \hat{\mathbf{r}} a d\theta.$$

Drop p_∞ piece ($\int \hat{\mathbf{r}} d\theta = 0$):

$$\mathbf{F} = 8\rho a^3 \beta^2 \int_0^{2\pi} \sin^2(2\theta + \alpha) \hat{\mathbf{r}}(\theta) d\theta.$$

Use $\sin^2(2\theta + \alpha) = \frac{1}{2}[1 - \cos(4\theta + 2\alpha)]$:

$$\mathbf{F} = 4\rho a^3 \beta^2 \int_0^{2\pi} [1 - \cos(4\theta + 2\alpha)] \hat{\mathbf{r}}(\theta) d\theta.$$

Both Fourier integrals vanish ($\int \hat{\mathbf{r}} d\theta = 0$; $\int \cos(4\theta + 2\alpha) \hat{\mathbf{r}} d\theta = 0$ as $\hat{\mathbf{r}} \propto e^{\pm i\theta}$ has no 4θ component):

$$\boxed{\mathbf{F} = \mathbf{0} \quad (\text{d'Alembert's paradox}).} \quad (18)$$

(d) Joukowski mapping and stagnation points

Mapped potential. Inverse map (branch $z \rightarrow Z$ as $|Z| \rightarrow \infty$):

$$z(Z) = \frac{1}{2}(Z + \sqrt{Z^2 - 4c^2}).$$

Far-field: $z \sim Z$, so the cylinder potential $w(z)$ pre-composed gives the required ellipse flow with $W \sim \beta Z^2 e^{i\alpha}$ at infinity. Hence

$$\boxed{W(Z) = \beta e^{i\alpha} [z(Z)]^2 + \frac{\beta a^4 e^{-i\alpha}}{[z(Z)]^2}, \quad z(Z) = \frac{1}{2}(Z + \sqrt{Z^2 - 4c^2}).} \quad (19)$$

Stagnation points. Chain rule:

$$\frac{dW}{dZ} = \frac{dw/dz}{dZ/dz}.$$

$dZ/dz = 1 - c^2/z^2$ vanishes at $z = \pm c$, both inside $|z| < a$ (since $c < a/2$): outside the physical domain. Zeros of dw/dz : factor (16):

$$z^4 = a^4 e^{-2i\alpha}.$$

Roots:

$$z_n = a e^{i(n\pi - \alpha)/2}, \quad n = 0, 1, 2, 3.$$

These lie on $|z| = a$, on the cylinder surface. Map by $Z = z + c^2/z$ with $\theta_n \equiv (n\pi - \alpha)/2$:

$$Z_n = a e^{i\theta_n} + \frac{c^2}{a} e^{-i\theta_n}.$$

Real and imaginary parts:

$$\boxed{Z_n = \left(a + \frac{c^2}{a}\right) \cos \frac{n\pi - \alpha}{2} + i \left(a - \frac{c^2}{a}\right) \sin \frac{n\pi - \alpha}{2}, \quad n = 0, 1, 2, 3.} \quad (20)$$

The four stagnation points lie on the ellipse with semi-axes $(a + c^2/a), (a - c^2/a)$. Limit $\alpha \rightarrow 0$: tips of the ellipse on its principal axes, as required by symmetry.